

This program is funded by the Australian Government Department of Foreign Affairs and Trade's Government Partnerships for Development Program and it will be delivered in Timor Leste from March 2014 to June 2016. The total funding for the program is \$1.8 million of which the Department of Foreign Affairs and Trade contributes \$1.4 million.

The program's main objective is to demonstrate effective poultry management strategies in a small number of "pilot villages" in Timor Leste. Management and vaccination techniques for Newcastle Disease demonstrated to be effective in improving village poultry production in Africa will be adapted to the Timor Leste pilot villages. Useful techniques demonstrated through the pilot villages will be communicated to other villages in Timor Leste using a targeted extension program.

The health of village poultry will also be protected by work to strengthen international border and village biosecurity to reduce the reintroduction of poultry diseases into Timor Leste and its villages. The program will be a partnership between the Timor Leste Ministry of Agriculture and Fisheries (MAF), the Australian Government Department of Agriculture and Water Resources and the University of Sydney's Veterinary School. This biosecurity work will promote better understanding of the importance of animal health at the village level in Timor Leste to improve poultry production outcomes.

The program supports both Timor Leste and Australian biosecurity interests. For Timor Leste the program provides a practical demonstration to villagers the direct benefit of better biosecurity on poultry health, welfare and production and improved food supply. Improved collaboration between villages and government will support more effective disease surveillance systems. Increased village poultry production will also support improved nutrition and prosperity for villagers.

Background

A large percentage of the Timor Leste population are dependent on rural primary production. Food and nutrition security is a continuing major challenge for Timor Leste. Subsistence agriculture is carried out by up to 80% of the population, marketable surpluses are not always produced and certain times of the year are marked by food scarcity in many villages. Timor Leste has high rates of malnutrition and child stunting (Timor Leste is severely affected according to UNICEF current statistics for this nutritional index). Increases in poultry production potentially will benefit some of the poorest members of the community through increased availability of high quality protein from eggs and chicken meat.

Poultry are found in almost all villages in Timor Leste. Poultry are generally managed with minimal inputs – birds search for scraps of food around the village. Poultry are consumed during special occasions or sold for cash – eggs are not generally utilised due to availability and cost. Disease is thought to be a major limiting factor impacting village poultry production.

Newcastle disease regularly causes high village chicken mortalities. The impact of disease is such that poultry numbers are kept at a level resulting in un-utilised food scraps in the village. In similar situations in village poultry in Africa effective Newcastle disease vaccination has demonstrated increased poultry production. A greater surety with regard to poultry survival also encourages greater human inputs into poultry production as returns are more certain. Effective Newcastle disease

vaccination is achieved in the village situation by vaccination of 75 % of the village chicken flock every 4 months by administering a live vaccine by eye drop. There are a number of challenges associated with achieving this level of vaccination including; proving to villages the advantages of vaccination, sustainability of vaccine supply, maintaining the vaccine cold chain, catching 75% of free running village chickens and repeating every 4 months.

While Newcastle disease vaccination has been widely practiced in Timor Leste for many years, it is thought to have been ineffective because the vaccine is not administered to a high enough percentage of the village poultry flock often enough to build up the flock immunity sufficiently to stop circulation of the virus. Maintaining an effective cold chain has been a challenge as power outages are common.

The improvement of other aspects of poultry production, such as biosecurity strengthening, better management of birds and increased utilisation of eggs, will lead to additional benefits.

A crucial aspect of this program is engagement with a small number of "pilot" villages to increase their knowledge of the benefit of effective Newcastle disease vaccination and to demonstrate the benefits of vaccination when regularly applied to a high percentage of the village flock.

Overview of the Timor Leste Village Poultry Health and Biosecurity Program

The program will identify management techniques that will increase the production of village poultry. The initial focus for production improvement will be through control of Newcastle disease by vaccination. A program customised to each location will identify the strategies that work best.

These strategies will be implemented, supported, and then gradually handed over to the villagers to maintain with support from MAF. A program of extension will then spread useful techniques to other villages in Timor Leste.

Village and international border biosecurity strengthening will support these strategies. The program will assist MAF in strengthening its biosecurity policy and operations at the international border to reduce the risk of introducing poultry diseases.

Women in developing countries play an important role in village poultry production. Engagement with this demographic of the village is important for the success of this program. Sydney University will partner with an international expert in social anthropology to better understand roles of men and women in the village, particularly:

- how men and women interact in Timor Leste society
- the place of poultry production in village priorities
- the main village demographic involved in poultry care
- who makes decisions about poultry utilisation and care, and
- what village communication strategies will be effective.

This information will ensure the Program is able to operate effectively in the village environment while supporting an agenda of promoting the involvement of women. Village heads will be encouraged to involve all sectors of the community with a role in poultry production including disabled villagers.

Program outcomes

- By the end of the program, vaccination and management techniques to increase small holder poultry production have been demonstrated in three pilot villages.
- Poultry production has increased in pilot villages (increased poultry numbers and production of eggs).
- Pilot villages understand the benefits and continue to implement successful poultry raising strategies suited to local conditions.
- MAF extension officers have been trained in the successful poultry raising strategies demonstrated in the program's pilot villages.
- MAF extension activities are starting to disseminate this information and provide technical support to other villages in Timor Leste (including mentoring of community vaccinators).
- Pilot villages have greater knowledge of biosecurity techniques that can protect the health of their poultry.
- MAF has improved knowledge on conducting an effective vaccination program for Newcastle disease.
- MAF has developed a draft annual budget for conducting an effective vaccination program for Newcastle disease in Timor Leste.
- Biosecurity policy has been reviewed and refined to better address poultry health risks posed by movement of high risk goods through the land border with Indonesia. These policies have been linked to village biosecurity practices.

Source: <https://www.agriculture.gov.au/animal/health/timor-leste-village-poultry>